

MD. KAMRAN HASAN RAFIN

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EXECUTIVE SUMMARY

Computer Science & Engineering undergraduate with a strong foundation in Applied Deep Learning, Explainable AI (XAI), Computer Vision and Software Architecture. Demonstrates cross-functional expertise spanning enterprise IT operations, full-lifecycle model training, algorithmic efficiency and network telemetry. Adept at transforming complex theoretical constructs into production-grade systems while applying rigorous root-cause analysis and performance benchmarking. With a particular interest in computer networking backed by coursework and hands-on experience in protocol analysis and network simulation.

CORE COMPETENCIES & TECHNICAL CAPABILITIES

Deep Learning & Computer Vision | Explainable AI (Grad-CAM, SHAP) | Edge AI & Model Optimization | Scalable Software Engineering (OOP, JavaFX) | Network Infrastructure & Protocol Diagnostics | Algorithmic Complexity Optimization | Enterprise IT Operations | Git & Github

EDUCATION

B.Sc. in Computer Science and Engineering / *Southeast University, Dhaka* **2022 – Present**

Academic Standing: CGPA 3.45 / 4.00 (Completed 122 of 150 Credits)
Core Coursework: Data Structures & Algorithms, Object-Oriented Software Design, Advanced Network Architecture, Communication Theory, Database Management Systems (DBMS).

FEATURED RESEARCH & PUBLICATIONS

PlantVisionBD: Explainable Deep Learning Framework for Multi-Crop Diagnostics / **Manuscript Complete**

- Architecture Engineering:** Engineered and deployed an EfficientNetV2L classification pipeline on an 87,800+ image dataset encompassing 38 disease categories across 14 plant species, achieving 98.43% test accuracy, 0.9991 Macro AUC, and a 12.4ms per-image inference speed.
- Optimization & Debugging:** Implemented a 3-phase progressive fine-tuning strategy with GeM pooling, Mixup/CutMix regularization, and EMA weight averaging. Identified and eliminated 9 critical pipeline bottlenecks, including a silent double-normalization gradient saturation defect.
- Explainable AI (XAI) & Interpretability:** Integrated Grad-CAM and SHAP saliency maps to mathematically validate lesion localization against background noise, conducting rigorous failure analysis on high-entropy edge cases.

PROFESSIONAL EXPERIENCE

IT Executive Officer / *Suffix IT Ltd. (Collaborative Enterprise Project with Abul Khair Tobacco Ltd.)* **2025**

- Operational Leadership:** Administered mission-critical IT operations, orchestrating system stability, access protocols and uptime monitoring across enterprise environments.
- Stakeholder & Cross-Functional Alignment:** Facilitated technical alignment between engineering personnel and non-technical management ensuring accelerated resolution cycles for high-priority operational tickets.
- Infrastructure Reliability & Root-Cause Analysis:** Executed diagnostic protocols to isolate hardware/network bottlenecks mitigating service interruptions and establishing comprehensive operational runbooks.

KEY ENGINEERING PROJECTS

- Exam Management System (Java, JavaFX, OOP):** Architected an event-driven desktop suite featuring dynamic assessment scheduling, role-based access control and decoupled state management via robust OOP principles.
- Student Management System (Java):** Designed a high-throughput console-based record management system using clean architectural patterns and persistent data handling structures.
- Algorithmic Performance & Optimization Suite (C++):** Implemented dynamic programming algorithms (0/1 Knapsack, optimal resource distribution) with comprehensive profiling of time complexity, memory allocation and CPU cache efficiency.

- **Network Infrastructure & Protocol Simulation (Cisco):** Modeled enterprise multi-tier topologies, evaluating throughput behavior, routing convergence, packet loss and latency under synthesized load constraints.

TECHNICAL SKILLS & TOOLING

- **Languages:** Python , Java, C++
- **AI, ML & Data Engineering:** TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Grad-CAM, SHAP, OpenCV, Data Augmentation Pipelines
- **Emerging & Futuristic Domains:** Explainable AI (XAI), Edge AI & Model Pruning, Network Automation, Distributed Telemetry
- **Networking, DevOps & Tools:** Cisco Packet Tracer, Protocol Analysis (TCP/IP, OSPF, BGP), Git, GitHub, VS Code, Jupyter, JavaFX

LANGUAGES & EXTRACURRICULAR LEADERSHIP

- **Languages:** Bangla (Native), English (Fluent - Professional Working Proficiency)
- **Leadership & Interests:** Open-Source AI Community Member, Mountain Trekking (Bandarban), Landscape & Travel Photography , Travelled hole Bangladesh